Long-Horizon State Tracking with Compact Mamba

Retention, Interference, and a Falsified Key-Routing Hypothesis

HackerRank Deep Learning Research Engineer Assignment · Primary architecture comparison: seed 42 · Supplementary diagnostics: seeds 42–46 · October 2026

Abstract. We investigate how a compact one-layer Mamba preserves, overwrites, and retrieves eight logical key–value registers. The trained baseline reaches 69.64% over all queries in length-512 sequences, yet controlled balanced retrieval falls from 84.38% at delay 10 to chance at delay 500. Unrelated-key SET events are more damaging than matched NOOP or QUERY controls. We therefore test key-conditioned write routing against a parameter-matched tied-row ablation. Their nearly identical accuracy and routing geometry falsify key selectivity as the source of the small observed gains. Mechanistic follow-ups instead identify accumulated passive transitions as the primary bottleneck: after the width-four convolution has washed out, forcing later NOOP recurrence to identity restores about 97% accuracy through delay 128. Under altered decay, a separately trained long-horizon linear readout can recover information that the original short-trained classifier cannot, exposing a secondary readout-distribution mismatch.

1. Architecture and Task

The task contains eight keys, four assigned values, and three event families: `SET(k,v)`, `QUERY(k)`, and `NOOP`. Its 41 logical tokens are 32 SETs, eight QUERYs, and one NOOP; the embedding table is padded to 48 entries. Each query predicts one of five classes {UNSET, 1, 2, 3, 4}. The QUERY input contains only the key, never its answer. Targets at non-query positions are masked with `-100`, so only queries contribute to cross-entropy.

The fixed backbone has d_{model}=32, one Mamba block, d_{state}=8, expansion 2 (d_{inner}=64), and a causal depthwise convolution of width four. A Linear(32,5) head produces logits. For convolution-processed input u_t, the readable scan is:

$$h_t = \exp(\Delta_t A) \odot h_{t-1} + \Delta_t B_t u_t, \quad r_t = C_t h_t + D \odot u_t, \quad A = -\exp(A_{\log}) < 0.$$

The state has 64×8=512 coordinates; Δ_t, B_t, and C_t are input-dependent. Retention depends on repeated multiplication by $\exp(\Delta_t A)$, while an unrelated event can both attenuate existing state and add a competing write. The causal convolution and left-to-right scan prevent future leakage. Thirty-three automated tests cover exact oracle semantics, UNSET and overwrite behavior, masking and leakage, shapes, finite gradients, causality, and routing behavior; all pass.

2. Hypotheses and Protocol

H1 — retention degradation. Increasing intervening NOOPs reduces balanced QUERY accuracy. It is falsified if long-delay accuracy is indistinguishable from short-delay accuracy. **H2 — cross-key interference.** At fixed delay, unrelated-key SETs reduce accuracy more than matched NOOPs or unrelated QUERYs. **Part B mechanism.** If H2 is caused by keys writing into overlapping recurrent coordinates, key-conditioned routing should beat a parameter-matched key-agnostic router by at least two percentage points under dense interference.

The supplied data contain 4,000 training and 500 validation sequences of lengths 32–128, plus five fixed 500-sequence splits: ID-128, OOD-256, OOD-512, delay-stress-512, and overwrite-stress-512. Training uses random initialization, FP32 AdamW, learning rate 10^{-3}, and batch size 16. The required 300-update model was undertrained, so the same model and optimizer were continued to 1,500 updates, selected by validation accuracy only. Baseline, proposed model, and ablation receive equal 1,500-update budgets. H1 uses balanced probes over seeds 42–44. H2 uses five independently trained baselines, 1,024 paired examples per condition, and matched active-token controls.

3. Baseline Results and Failure Diagnosis

Table 1. All-query accuracy (%) on fixed assignment splits. The exact dictionary oracle is 100% throughout.

Split	Length	300 updates	1,500 updates	Always UNSET
ID	128	25.67	**75.05**	23.28
OOD	256	14.81	**71.23**	11.28
OOD	512	10.45	**69.64**	5.97
Delay stress	512	8.64	**69.59**	4.21
Overwrite stress	512	8.86	**69.67**	4.28

Aggregate length-512 accuracy conceals dependency age: many queries concern recently written values. In the balanced H1 test, mean accuracy is 84.38, 64.58, 45.83, 28.13, 23.96, and 25.00% at delays 10, 32, 64, 128, 256, and 500. The paired drop from delay 10 to 128 is 56.25 points (95% seed-level CI 40.72–71.78); delay 500 is at four-value chance. H1 is supported.

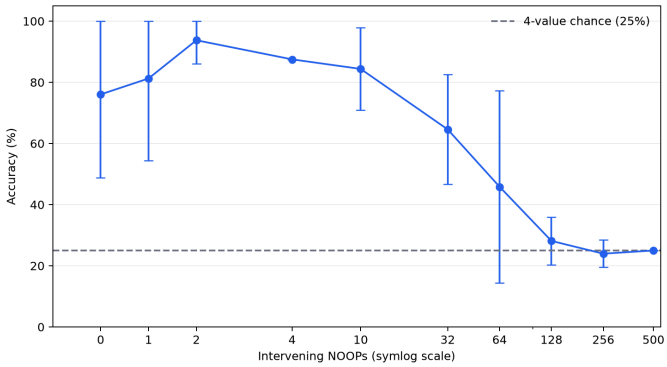


Figure 1. Balanced retrieval degrades with intervening NOOPs across three training seeds. Wide early intervals reflect n=3; convergence toward chance is consistent across seeds.

At delays 10/32/64/128, NOOP-only accuracy is 93.16/82.23/54.16/29.47%; replacing every intervening slot with an unrelated SET gives 78.67/31.74/21.80/21.21%. The SET-minus-NOOP effects are −14.49, −50.49, −32.36, and −8.26 points. Unrelated QUERY controls are less damaging at intermediate delays, ruling out a simple "any active token" explanation. Delay 128 is floor-limited rather than evidence of recovery.

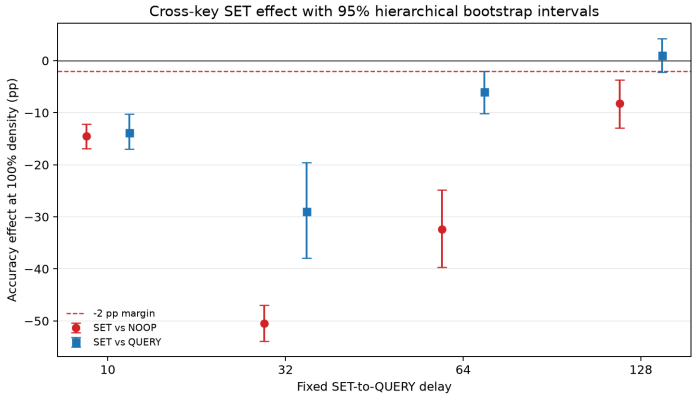


Figure 2. Five-seed cross-key effects at 100% distractor density. Negative values mean unrelated SETs are worse than the matched control; the dashed line is the predeclared −2-point practical margin.

4. Part B: Key-Conditioned Write Routing

The proposed model learns a routing table R with shape 8×64. For SET to key k, $g_k=\sigma(R_k)$ gates only the input-write contribution:

$$h_t = \exp(\Delta_t A)\odot h_{t-1} + g_k\odot(\Delta_t B_t u_t).$$

This adds 512 parameters (10,661 total, +5.0%). Non-SET writes are unchanged. The tied-row ablation keeps the same parameter table but averages rows during computation, eliminating key selectivity while retaining capacity. Both initialize to gate value 0.95.

Table 2. Matched 1,500-update models: all-query accuracy (%).

Model	ID-128	OOD-256	OOD-512	Delay	Overwrite
Baseline	75.05	71.23	69.64	69.59	**69.67**
Key-conditioned	**75.57**	**71.54**	**69.86**	**70.06**	69.55
Tied routing	75.50	71.42	69.75	69.96	69.50

The proposed advantage over the matched ablation is only 0.05–0.12 points on fixed splits. At 100% unrelated-SET density, key-conditioned-minus-tied effects at delays 10/32/64/128 are +0.49, +0.10, −0.39, and +0.29 points; none approaches the two-point criterion. Learned rows are nearly identical (mean cosine similarity 0.99958; mean normalized row distance 0.0286; mean gate 0.946), indicating global write rescaling rather than key-specific partitions.

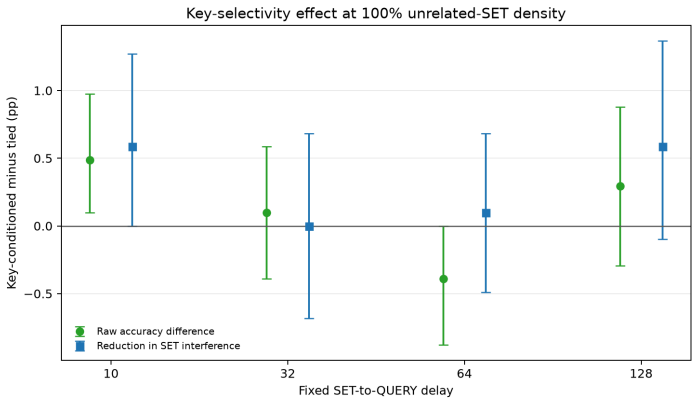


Figure 3. Mechanism-specific Part B contrast. Intervals remain centered close to zero: key selectivity does not reliably outperform the parameter-matched tied router.

Final-query overwrite accuracy for 1/2/4/8 focus-key overwrites is 88.8/88.0/82.4/90.4% with baseline, 90.4/87.2/80.8/91.2% with key routing, and 90.4/87.2/80.8/92.0% with tied routing. Stale-value errors are 8.6, 9.0, and 8.8%. The non-monotonic curve means overwrite count is not an isolated difficulty variable. The Part B mechanism is falsified for this run: small gains are explained by shared routing capacity, not key identity.

5. Mechanistic Follow-up

Write routing leaves global attenuation untouched. I therefore measured NOOP state displacement and performed two frozen interventions. Forcing identity immediately after SET destroys accuracy because the width-four convolution continues the write response for three steps. Allowing that washout, then forcing later NOOP recurrence to identity, yields 96.84/96.70/97.05/96.60% at delays 10/32/64/128 across five seeds, versus 93.36/81.15/54.47/29.32% normally. This is the largest causal rescue observed and supports harmful accumulated late-NOOP transitions as the primary bottleneck. It remains an oracle, out-of-distribution intervention—not a replacement architecture.

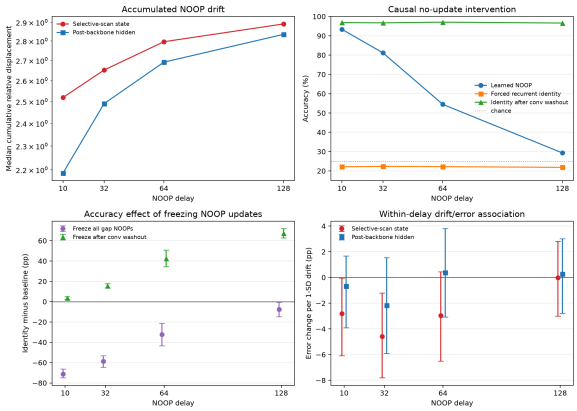


Figure 4. NOOP identity diagnostic. Immediate freezing removes necessary convolutional settling, whereas freezing only after three washout steps preserves near-97% retrieval. Raw drift alone is not sufficient evidence; the intervention is discriminating.

A separate frozen-backbone test varies only post-washout decay over a 511-step gap and independently trains linear probes. With zero late decay, the original task head scores 25.8%, while a probe on the same final 32D feature reaches 99.6%. A fresh head trained only on lengths 32–128 remains near chance, so initialization or optimizer choice is not the explanation; long-horizon feature supervision is decisive. At natural decay, even the long-trained final-feature probe reaches only 39.5%, showing that readout adaptation alone cannot repair natural recurrence. Probe scores measure linear decodability and are not fair replacement-model results.

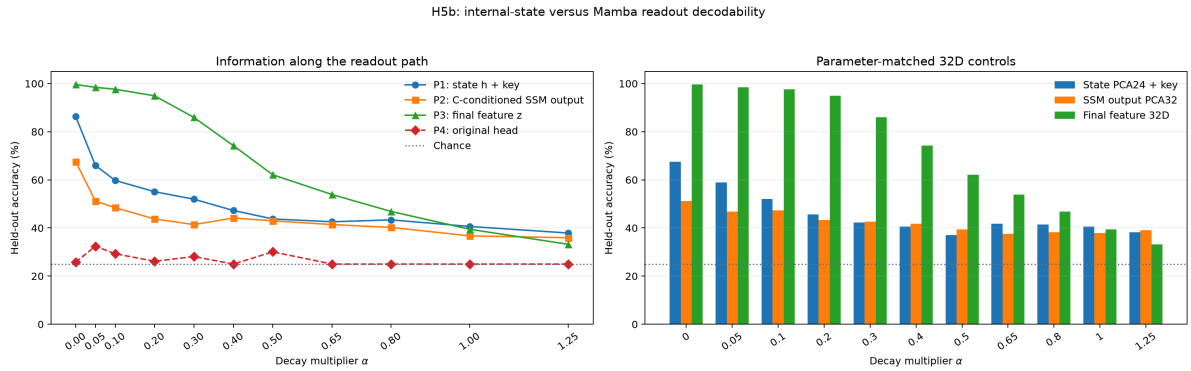


Figure 5. Held-out probes along the readout path. Under reduced decay, the final feature is highly decodable by a matched long-horizon probe while the original head remains near chance; under natural decay (α=1), accessibility is much lower throughout.

6. Failure Analysis, Limitations, and Next Step

The evidence supports three bounded conclusions. **First**, natural long-horizon retention is primary: controlled accuracy collapses with delay and the convolution-aware identity intervention produces a large rescue. **Second**, unrelated-key SETs create real interference, but gating only their writes is insufficient because shared decay and projections remain. **Third**, altering retention creates a feature distribution that the original short-trained head does not interpret reliably. Preservation and retrieval must therefore be co-adapted.

The Part B architecture comparison uses one training seed, so sub-point optimization effects remain uncertain. This does not rescue the mechanism claim: effects are far below the practical threshold and routing rows fail to specialize. H1 uses only three model seeds, producing wide intervals. Frozen interventions and long-trained probes deliberately change deployment assumptions and establish mechanisms, not benchmark improvements. The readable scan is unoptimized, so timing (449.6 s baseline, 953.6 s keyed, 1,162.3 s tied) should not be extrapolated to fused kernels. Peak allocation is 50.7 MiB for baseline and 55.2 MiB for routed models.

Single most informative next experiment. Train a convolution-aware no-update gate: leave the first $d_{\text{conv}}-1=3$ post-SET transitions unconstrained, then let later NOOP steps interpolate between ordinary recurrence and identity. Train only on lengths 32–128 and compare baseline, event-aware gating, and a parameter-matched event-agnostic gate over at least five seeds. Primary outcomes are untouched final-query accuracy at delays 128/256/511, immediate overwrite accuracy, stale-value error, interference, parameters, runtime, and memory. Falsify the proposal if it stabilizes short-horizon geometry without improving delay 511, or if retention gains create a material overwrite penalty.

7. Reproducibility and Attribution

Fixed splits, predictions, checkpoints, intervals, and plots are stored under `results/`; detailed exploratory analyses remain separate from this focused report. Core commands use `train.py` with `configs/baseline.yaml`, `configs/proposed.yaml`, or `configs/ablation.yaml`, followed by `part_b_compare.py`. Tests run with `python -m pytest -q -p no:cacheprovider` (33 passed).

The readable Mamba code derives from John Ma's MIT-licensed *mamba-minimal*; equations follow Gu and Dao's Mamba formulation. The dataset generator and sample data were supplied with the assignment. AI assistance was used substantially for scaffolding, experiment scripts, plotting, and editing. I inspected code paths, selected hypotheses and controls, executed experiments, verified artifacts, and take responsibility for the conclusions.

References. [1] A. Gu and T. Dao, "Mamba: Linear-Time Sequence Modeling with Selective State Spaces," arXiv:2312.00752, 2023. [2] J. Ma, *mamba-minimal*, MIT-licensed PyTorch implementation, github.com/johnma2006/mamba-minimal.

Gate-Free Adaptive-Kernel Diagnostic

All models use the original Mamba recurrence with no event-conditioned retention gate. Original Mamba is the fixed-width-4 control; fixed width 8 changes only the causal convolution. Static and adaptive mixtures share a maximum-width-8 kernel and identical 10,729-parameter capacity, but only adaptive mixture conditions its width weights on the current token representation. Every model uses seed 42 and 1,500 updates on lengths 32-128.

Model	ID-128	OOD-512	Gap-512	Overwrite-512	Final-gap
Original / width 4	75.05	69.64	69.59	69.67	22.8
Fixed width 8	76.59	71.94	71.54	71.86	22.6
Static mixture	77.62	72.94	72.61	72.56	24.4
Adaptive mixture	78.95	73.91	73.58	73.98	23.2

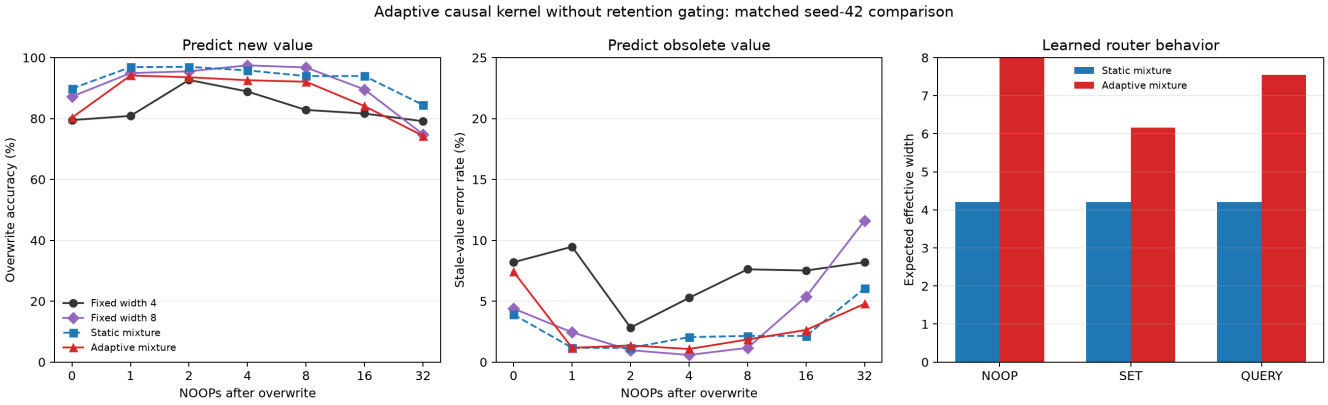


Figure 6. In the gate-free comparison, adaptive routing chooses wide contexts but static mixing has higher overwrite accuracy at every tested post-delay.

Assignment-Defined Stress Evaluation

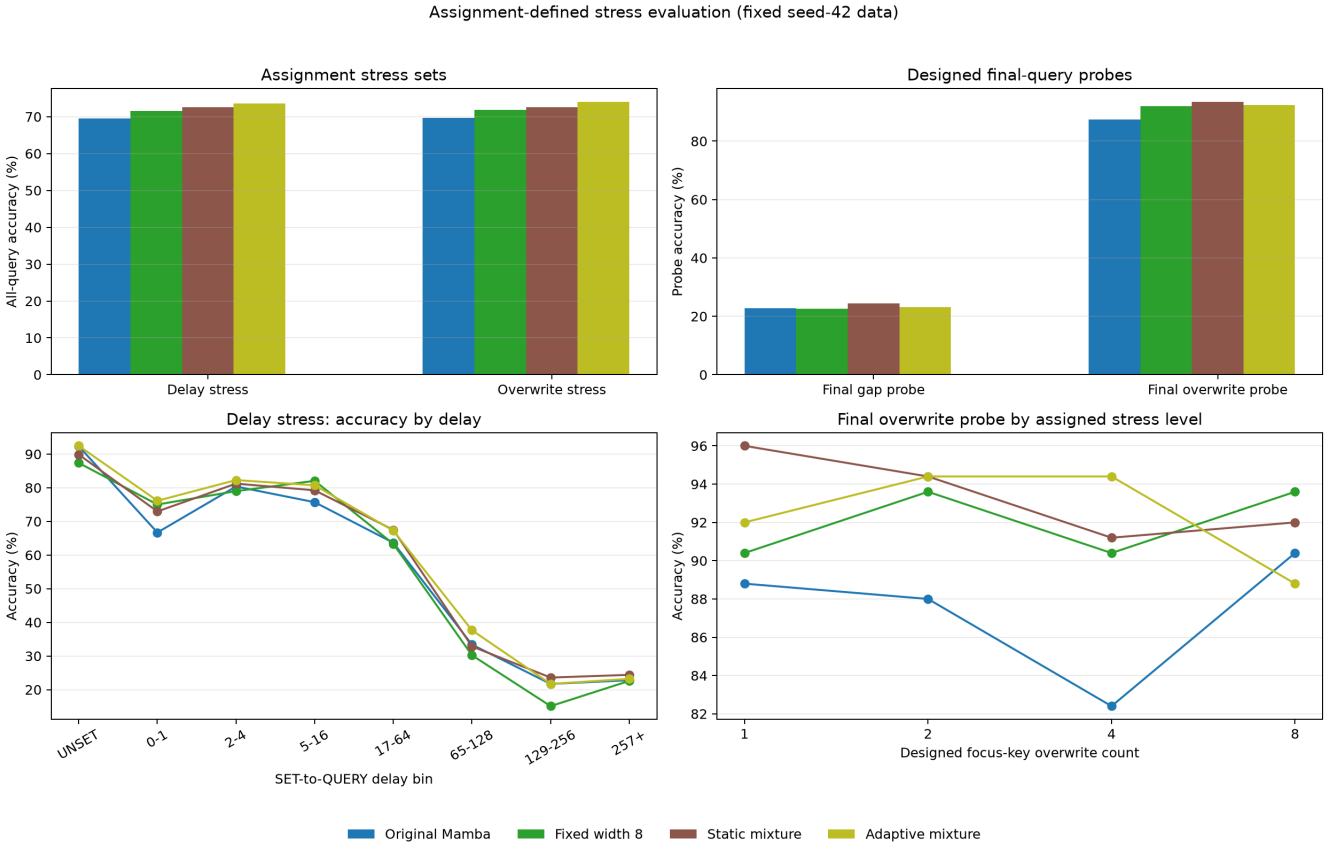


Figure 7. Adaptive mixing has the best aggregate stress accuracy, but final-gap retrieval remains near four-value chance and the static control is slightly stronger on both designed final probes.

Paired controlled probes. Adaptive versus static changes final-gap accuracy by -1.2 pp (95% CI -5.6 to 3.0) and final-overwrite accuracy by -1.0 pp (95% CI -3.2 to 1.2). Adaptive versus original improves overwrite by +5.0 pp (95% CI 1.2 to 8.8), but static reaches 93.4% versus adaptive's 92.4%.

Supplementary verdict. Token-conditioned convolution is not supported over parameter-matched static mixing. Its aggregate gains do not transfer to the controlled final queries, and extreme retention remains unsolved. These seed-42 results are supplementary and do not replace the mandatory baseline or Part B comparison.

Training-Time and Memory Trade-off

All measurements use 1,500 updates in FP32 on the same NVIDIA GeForce GTX 1650. Labels report the recorded wall-clock training time and peak PyTorch GPU allocation.

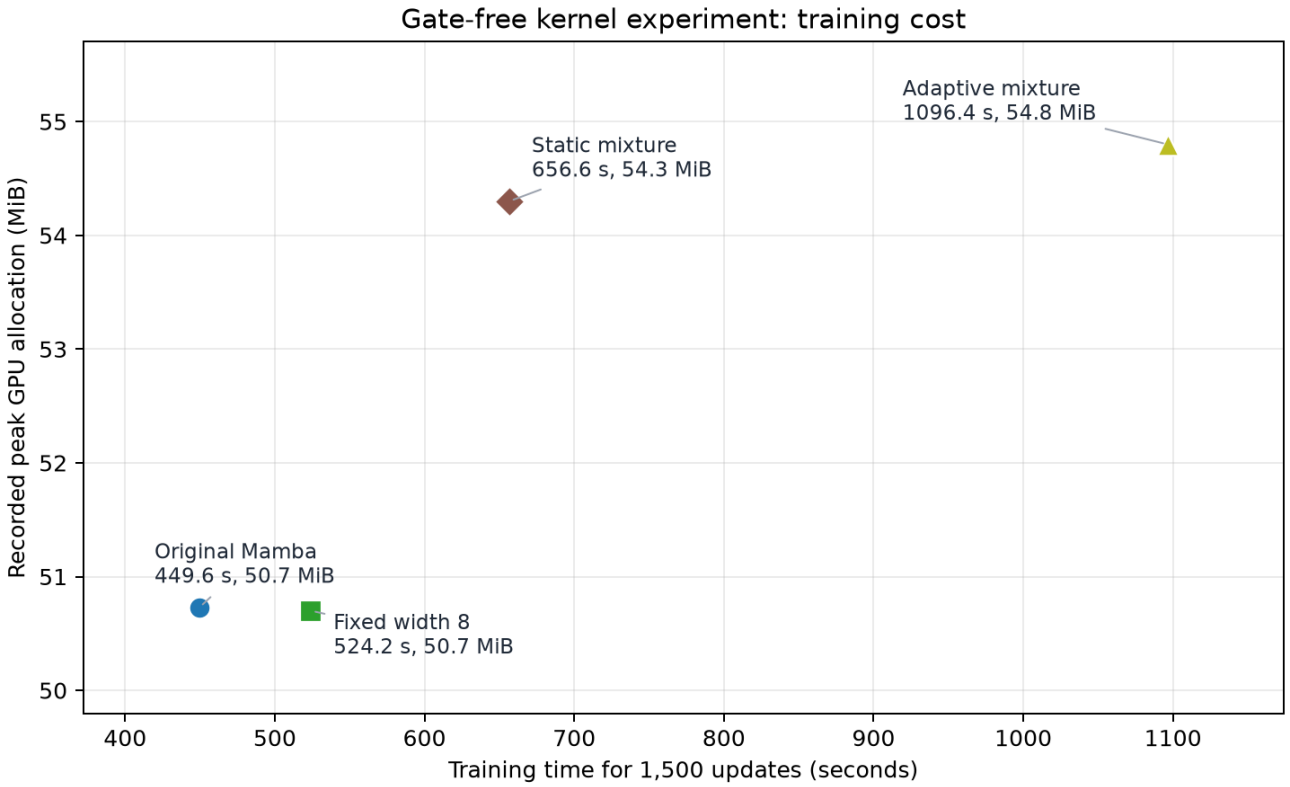


Figure 8. The adaptive router is substantially more expensive in time but only modestly more expensive in GPU memory.

Engineering takeaway. Adaptive mixing takes 1,096.4 s, or 2.44 times the original Mamba and 1.67 times the static mixture. Peak allocation is 54.8 MiB, only 4.1 MiB above original and 0.5 MiB above static. In this readable implementation, token-dependent kernel construction is primarily a compute-time cost rather than a memory-capacity cost.